Mohamad Hassan Ibrahim

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Education

Telecom Paris, Institute Polytechnique de Paris — Île-de-France, France
Software Engineering — *Expected Graduation 2027*

Lebanese University, Faculty of Engineering III — Beirut, Lebanon
Computer and Telecommunication Engineering — *Expected Graduation 2027*

Experience

Software Engineering PRIM Project Intern — Oct. 2026 – Jan. 2027
Proxima — *Paris, France*

- Configured and secured internal company services under their VPN
- Performed penetration testing on LLM-based applications to identify security vulnerabilities and attack vectors
- Conducted web application penetration testing and documented identified security risks

Network Engineer Internship — Aug. 2024 – Sep. 2024
Broadband ISP Ferrari Networks, Member of Nova Holding — *Zefta, Lebanon*

- Troubleshot and resolved network issues across multiple sites using the MikroTik software
- Configured Mikrotik switches, routers, wireless dish and access points
- Installed networking devices using fiber optic and Ethernet cables on-site across various sites

Tutor and Board Member — Aug. 2024 - Aug. 2025
Cyber Community — *Lebanese University*

- Offered tutoring sessions for Networking, Python, and Cryptography Courses

Academic Tutor — Oct. 2022 - Dec. 2026
Faculty of Engineering III — *Lebanese University*

- Provided tutoring sessions in "Introduction to C Language", "Introduction to OOP in C#", "OOP and Design Patterns in C++", "Computer Networks" to engineering students

Projects

Padel & Tennis Court Reservation App | *React Native, Expo, TypeScript, Supabase, PostgreSQL* — Aug. 2026

- Developed a cross-platform mobile application for padel and tennis court reservations with authentication, user profiles, and dynamic court availability
- Built a reservation system supporting multiple booking durations, overlap prevention, court-specific pricing, blocked time slots, and real-time availability updates
- Implemented role-based administration with protected navigation and Supabase Row Level Security, enabling admins to manage announcements, posts, and reservation-related content

Passive Haptics and Movement Gain in Virtual Reality | *Unity, C#, OpenXR, XR Hands* — June 2026

- Designed and developed a VR interaction system combining passive haptics and movement gain to make a single physical board behave as multiple virtual objects
- Implemented the Byblos puzzle scene and Maze Game, using precise fingertip-based interaction for continuous surface navigation and tactile feedback
- Developed fixed gain behavior to amplify real hand movement in VR, allowing users to reach virtual elements while remaining within a limited physical space
- Collaborated on calibration, virtual surface alignment, gain testing, interaction design, visual scene assembly, report writing, and final presentation preparation

Geo-Social Dating App | *React Native, AWS Cognito, Lambda, PostgreSQL, S3, ChatGPT API* — April 2026

- Developed a geo-social dating app with a frontend in React Native/Expo and a scalable serverless backend on AWS
- Built backend services for authentication, user profiles, location-based discovery, matching, real-time messaging, posts, stories, and media storage
- Implemented a ranking algorithm combining interests, distance, and introversion similarity for personalized user discovery

- Added AI assistant and chat suggestion features with the ChatGPT API, plus profile verification using AWS Rekognition

Fundamentals of Computer Graphics Project | *OpenGL, Blender, Ray Tracing, BVH* Jan. 2026

- Built a hybrid rasterization and ray-tracing renderer accelerated with BVH optimization
- Applied vertex welding and bilateral smoothing to improve a decimated 3D model
- Built a particle-based waterfall simulation and integrated animated droplets into the ray-traced output

Level 1 SOC Analyst AI Agent | *Python, Langchain, FastAPI, React, PostgreSQL, Docker* July 2025

- Built a Dockerized SOC AI Agent prototype for automated alert triage
- Integrated external tools using API request methods for threat enrichment
- Designed JSON playbooks for common attack scenarios
- Developed backend (FastAPI + PostgreSQL) and frontend dashboard for alert reports

Multi-threaded Cut Scene Detector | *Java, OpenCV* June 2025

- Designed a cut scene detection system in Java, leveraging multi-threading
- Conducted a comparative analysis of single-threaded, producer–consumer, and parallel processing models
- Evaluated performance trade-offs to demonstrate efficiency gains through concurrency

Complete Campus Area Network | *Packet Tracer, CCNA* Sep. 2024

- Designed and configured a complete campus area network for buildings using several technologies:
  - VLANs, Inter-VLAN routing and OSPFv2
  - Firewall security, ACL configurations, and IPsec VPN

Computer Vision Drone Detection Model | *Python, YOLOv8, vast.ai* June 2024

- Fine-tuned YOLOv8 for drone detection, optimizing model size for speed and accuracy
- Created a custom drone dataset using personal and online images

Certificates

NetAcad CCNA1 course | Introduction to Networking *Issued in Feb. 2024*
NetAcad CCNA2 course | Switching, Routing, and Wireless Essentials *Issued in July 2024*
NetAcad CCNA3 course | Enterprise Networking, Security, and Automation *Issued in Feb. 2025*

Technical Skills & Languages

Programming: Java/C#, Python, C/C++, SQL (PostgresSQL & mySQL), JavaScript/Typescript
Frameworks: React js, Node.js, Three.js, FastAPI, Springboot, Langchain, Qt, OpenGL, React Native
Developer Tools: Git / GitHub, Docker, Blender, SFM, Unity, Supabase, Brevo
Languages: Arabic (Native), English (IELTS 7.0 - C1), French (Intermediate)